Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 23)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

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0 1 . 2 Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

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0 1 . 3 Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NOx.

[3 marks]

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0 1 . 4 Explain why carbon monoxide is toxic to humans.

[2 marks]

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0 2 Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 2 . 1 Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

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0 2 . 2 Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

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0 2 . 3 Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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0 2 . 4 Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

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0 3 Table 1 shows the average annual radiation dose received by a person in the UK from different sources.

Table 1

Source	Percentage of average dose / %
Radon gas from the ground	50
Medical (X-rays, CT scans)	16
Gamma rays from rocks / buildings	13
Internal (food and drink)	9.5
Cosmic radiation	10
Occupational, fallout, nuclear discharges, consumer products	1.5

0 3 . 1 Explain why radon concentrations are high inside some houses in Devon and Cornwall.

[3 marks]

0 3 . 2 Use Table 1 to calculate the percentage of the average dose that comes from artificial (human-made) sources.

[1 mark]

Answer ☐ %

0 3 . 3 Suggest **two** reasons why the dose received by an individual may differ greatly from the average.

[2 marks]

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0 3 . 4 Explain why a risk:benefit analysis is needed for the use of X-rays in medicine, and give **two** reasons why the risks of ionising radiation are difficult to assess.

[4 marks]

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0 4 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 4 . 1 Explain what the standard deviation values tell you about the data.

[2 marks]

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0 4 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

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0 4 . 3 The test gave a probability p = 0.30. State what this means and what the student should conclude.

[2 marks]

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0 4 . 4 Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

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0 4 . 5 Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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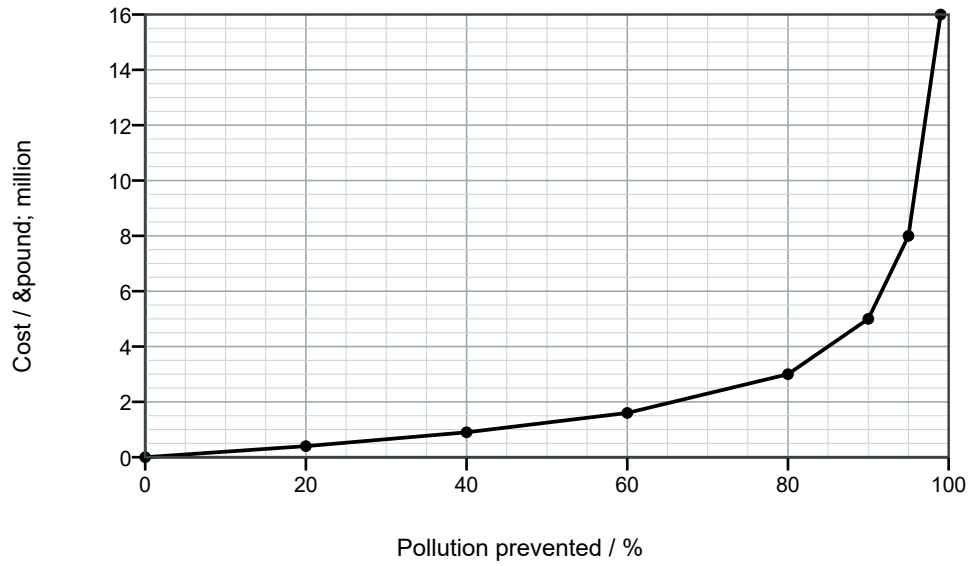
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0 5 Figure 1 shows the relationship between the cost of pollution control equipment at a factory and the percentage of a pollutant emission that is prevented.

Figure 1



0 5 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 5 . 2 Use Figure 1 to calculate the extra cost of increasing the pollution prevented from 90% to 99%.

[1 mark]

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Answer [ ] £ million

0 5 . 3 Explain what is meant by the terms **ALARA** and **BATNEEC** and how they would be applied by the factory.

[4 marks]

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0 5 . 4 Suggest why the control methods used originally by a factory may no longer be sufficient after several years.

[3 marks]

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0 6 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 6 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 6 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 7 A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

0 7 . 1 Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

[5 marks]

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0 7 . 2 Suggest **one** reason why such schemes may not be installed even at suitable sites.

[1 mark]

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0 7 . 3 The maximum flow rate at the site is 6.5 m^3 s^{-1} giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

[2 marks]

Answer ☐ kW

0 7 . 4 Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

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2 .....

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0 8 . 1 Explain why current global energy use is described as unsustainable.

[3 marks]

0 8 . 2 Describe the environmental impacts of exploiting fossil fuels that occur **before** the fuel is used.

[4 marks]

0 8 . 3 Explain why an 'energy gap' could occur if society waits until fossil fuels become expensive before developing alternatives.

[3 marks]

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0 9 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

0 9 . 1 Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

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0 9 . 2 Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage .....

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Disadvantage .....

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0 9 . 3 Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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0 9 . 4 Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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1 0 . 1 Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS